# Benchmarking of NQCH's quantum computer

*October 14, 2026*

## 1. Report of Changes

**Platform:** sinq20
**Calibration-id:** *ae51b7967ef92463a98d705f04fb00e9d9d10190*
**Calibration date:** *2026-08-12 02:52:08*
**Calibration note:** chore(sinq20): q5 sq recal

**Experiment-id:** *20261014212203*
**Experiment date:** *2026-10-14 21:22:04*
**Experiment note:** *temporary note!!!*

**Platform:** sinq20
**Calibration-id:** *4dc4082f38a53222b3956c22202d32a520d4bc78*
**Calibration date:** *2026-01-15 02:03:03*
**Calibration note:** chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

**Experiment-id:** *20260118161538*
**Experiment date:** *2026-01-18 16:15:38*
**Experiment note:** *temporary note!!!*

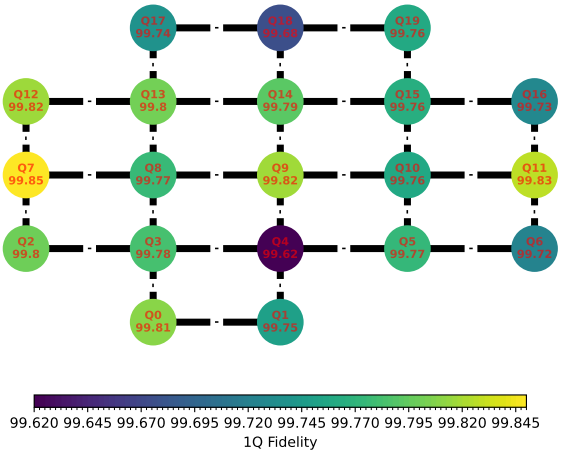
## 2. Version Comparison

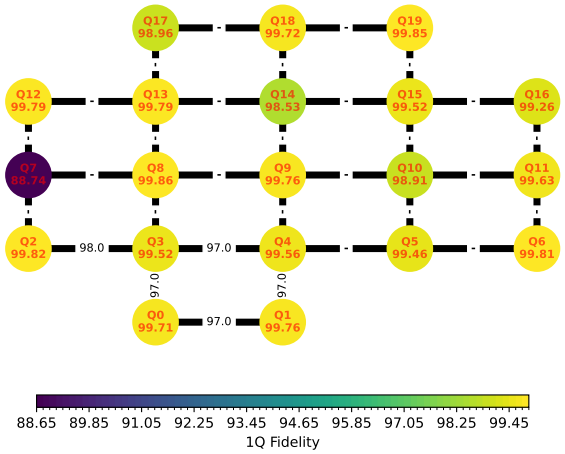
| Library | Version | Library | Version |
|---|---|---|---|
| qibo | 0.3.3 | numpy | 2.5.3 |
| qibolab | 0.2.17 | qibocal | 0.2.7 |
| matplotlib | 3.11.1 | scipy | 1.18.1 |
| scikit-learn | 1.9.0 | pandas | 3.0.5 |
| networkx | 3.6.1 | sympy | 1.14.0 |
| torch | 2.14.0 | | |

| Library | Version | Library | Version |
|---|---|---|---|
| qibo | 0.2.22 | numpy | 2.2.6 |
| qibolab | 0.2.9 | qibocal | 0.2.4 |
| matplotlib | 3.10.3 | scipy | 1.15.3 |
| scikit-learn | 1.6.1 | pandas | 2.2.3 |
| networkx | 3.4.2 | sympy | 1.14.0 |
| torch | 2.7.0 | | |

## 3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



## 4. Statistics

| | Average | Median | Min | Max |
|---|---|---|---|---|
| T1 (ns) | - | - | - | - |
| T2 (ns) | - | - | - | - |
| Fidelity | - | - | - | - |
| RO Fidelity | - | - | - | - |
| 2q RB Fidelity | - | - | - | - |
| 2q CZ Fidelity | - | - | - | - |

| | Average | Median | Min | Max |
|---|---|---|---|---|
| T1 (ns) | 1.28e+04 | 1.23e+04 | 646 | 3.65e+04 |
| T2 (ns) | 2.36e+25 | 3.68e+03 | 125 | 9.43e+26 |
| Fidelity | 0.99 | 0.997 | 0.887 | 0.999 |
| RO Fidelity | 0.794 | 0.777 | 0.777 | 0.927 |
| 2q RB Fidelity | 0.965 | 0.972 | 0.923 | 0.981 |
| 2q CZ Fidelity | 0.975 | 0.974 | 0.967 | 0.991 |

## 5. Best Qubits Selection

| k-qubits | Best Qubits | Fidelity |
|---|---|---|
| 2 | N/A | N/A |
| 3 | N/A | N/A |
| 4 | N/A | N/A |
| 5 | N/A | N/A |

| k-qubits | Best Qubits | Fidelity |
|---|---|---|
| 2 | 2, 3 | 0.981 |
| 3 | 0, 2, 3 | 0.976 |
| 4 | 0, 1, 2, 3 | 0.970 |
| 5 | 0, 1, 2, 3, 4 | 0.965 |

## 6. Summary metrics

| Benchmark | Metric | Value | Runtime |
|---|---|---|---|
| Mermin | Max | - | - |

| Benchmark | Metric | Value | Runtime |
|---|---|---|---|
| Mermin | Max | -3.316 | 1.02 sec |

## 7. Randomized Benchmarking Results

| Qubit n | Fidelity | Error Bars | Qubit n | Fidelity | Error Bars |
|---|---|---|---|---|---|
| 0 | 0.998 | ± 8.62e-05 | 10 | 0.998 | ± 0.000103 |
| 1 | 0.997 | ± 0.000103 | 11 | 0.998 | ± 6.63e-05 |
| 2 | 0.998 | ± 9.66e-05 | 12 | 0.998 | ± 6.58e-05 |
| 3 | 0.998 | ± 0.000153 | 13 | 0.998 | ± 0.000102 |
| 4 | 0.996 | ± 0.000226 | 14 | 0.998 | ± 8.09e-05 |
| 5 | 0.998 | ± 0.000124 | 15 | 0.998 | ± 0.000103 |
| 6 | 0.997 | ± 0.000121 | 16 | 0.997 | ± 0.000162 |
| 7 | 0.998 | ± 4.57e-05 | 17 | 0.997 | ± 8.82e-05 |
| 8 | 0.998 | ± 0.000107 | 18 | 0.997 | ± 0.000138 |
| 9 | 0.998 | ± 0.000117 | 19 | 0.998 | ± 8.11e-05 |

| Qubit n | Fidelity | Error Bars | Qubit n | Fidelity | Error Bars |
|---|---|---|---|---|---|
| 0 | 0.997 | ± 0.000131 | 10 | 0.989 | ± 0.000582 |
| 1 | 0.998 | ± 7.47e-05 | 11 | 0.996 | ± 0.000218 |
| 2 | 0.998 | ± 7.57e-05 | 12 | 0.998 | ± 0.000126 |
| 3 | 0.995 | ± 0.000163 | 13 | 0.998 | ± 7.15e-05 |
| 4 | 0.996 | ± 0.000185 | 14 | 0.985 | ± 0.00183 |
| 5 | 0.995 | ± 0.000147 | 15 | 0.995 | ± 0.000469 |
| 6 | 0.998 | ± 8.34e-05 | 16 | 0.993 | ± 0.000558 |
| 7 | 0.887 | ± 0.0287 | 17 | 0.99 | ± 0.000502 |
| 8 | 0.999 | ± 4.48e-05 | 18 | 0.997 | ± 0.00026 |
| 9 | 0.998 | ± 7.2e-05 | 19 | 0.998 | ± 9.05e-05 |

## 8. Mermin

| Runtime | Qubits | Max |
|---|---|---|
| - | - | - |



| Runtime | Qubits | Max |
|---|---|---|
| 1.02 sec | 0, 2, 3 | -3.316 |

